

WhatsApp integration

Setup guide for the Navin Plastic Tech CRM

The front door [BLUEPRINT §41]. Once this is wired up, a message a buyer sends to the plant's WhatsApp number lands in the CRM, is matched against the customer book, and appears in marketing's inbox already assigned to the right person.

This guide covers the **inbound** side — messages arriving. The outbound side (§42, *Send to customer*) shares the same Twilio account and is covered at the end.

Before you start

What you need	Notes
A Twilio account	twilio.com — free to open, then pay per message
A Facebook Business Manager account	Required by Meta for any production sender
The server reachable over HTTPS	Twilio will not post to an IP address or plain HTTP
Time	Ten minutes for the sandbox; a few days for Meta's verification

You do not need the WhatsApp Business app on a phone

In fact you must not use the same number in both — a number connected to Twilio stops working in the phone app.

1 · Try it on the Twilio sandbox first

The sandbox is a shared Twilio number anybody can message after sending a join code. It behaves exactly like the real thing for inbound messages and takes two minutes. Do this before starting the Meta paperwork, so that when the real number arrives you already know the plumbing works.

- 1 In the Twilio console, go to **Messaging** → **Try it out** → **Send a WhatsApp message**.
- 2 Note the sandbox number (usually +1 415 523 8886) and the join phrase, e.g. `join heavy-cloud`.
- 3 From your own phone, send that join phrase to the sandbox number on WhatsApp.

2 · Configure the server

Three values go in the server's `.env`. Two you already have if outbound messaging is set up.

```
# From Twilio console -> Account Info
TWILIO_ACCOUNT_SID=ACxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

TWILIO_AUTH_TOKEN=your-auth-token

# The WhatsApp sender. Sandbox number while testing, your own once approved.
# Note the `whatsapp:` prefix - it is part of the value.
TWILIO_WHATSAPP_FROM=whatsapp:+14155238886

# The inbound guard. Generate a long random string.
WHATSAPP_WEBHOOK_TOKEN=
```

Generate the token, then restart the API:

```
openssl rand -hex 32

pm2 restart npt-api    # or however the service is run
```

If `WHATSAPP_WEBHOOK_TOKEN` is unset, the webhook answers 503 and accepts nothing
That is deliberate. An unguarded inbound route is worse than a broken one.

3 · Point Twilio at the server

In the Twilio console go to **Messaging** → **Try it out** → **Send a WhatsApp message** → **Sandbox settings**.
For a real number: **Messaging** → **Senders** → **WhatsApp senders** → **your number**.

Set “**When a message comes in**” to the URL below, with method **HTTP POST**:

```
https://your-domain.com/api/whatsapp/inbound?token=THE_TOKEN_YOU_GENERATED
```

Why the token is in the URL

Twilio's webhook configuration takes a URL and a method and nothing else — there is no field for a custom header. The server accepts the token either as an `x-webhook-token` header or as a `?token=` query parameter for exactly this reason. **The URL is therefore a secret.** It goes in the Twilio console and nowhere else — treat it like a password, and keep it out of tickets, chat messages and screenshots.

Leave “**Status callback URL**” empty. That is for delivery receipts on outbound messages and is not used by the inbox.

Checking it

Send a WhatsApp message to the sandbox number from your phone. Then, signed in to the CRM as an admin:

```
curl -s https://your-domain.com/api/whatsapp/threads \
-H "Authorization: Bearer YOUR_TOKEN" | head -40
```

4 · Going live on a real number

The sandbox is fine for testing and useless for customers — they would each have to send a join code first. For production you need a WhatsApp sender approved by Meta.

- 1 **Choose a number.** It must be one that is *not* currently registered to WhatsApp. A number already in the WhatsApp Business phone app has to be deleted from there first, and that takes effect after a delay — plan for it. A fresh number is simpler.
- 2 In Twilio: **Messaging** → **Senders** → **WhatsApp senders** → **New sender**, and follow the prompts. Twilio walks you through connecting Facebook Business Manager.
- 3 **Business verification.** Meta verifies the business behind the sender — company name, address, website, and usually a utility bill or GST certificate. This is the slow part: budget a few days, occasionally a couple of weeks.
- 4 **Display name.** What customers see above the chat. Meta rejects names that do not match the verified business, so use the registered name — *Navin Plastic Tech* rather than a brand nickname.
- 5 Once approved, change `TWILIO_WHATSAPP_FROM` to the new number and restart. Move the webhook URL from the sandbox settings to the sender's own configuration.

5 · Give marketing access

The inbox is behind the what sapp module grant, like every other screen.

- **Admin** → **Users** → *person* → **Access**, and grant what sapp.
- `read` lets somebody see the inbox. `write` lets them assign, link and convert.
- Marketing people see **their own** conversations; management sees all of them [§29].

Converting a conversation into an enquiry additionally needs `enquiries: write` — the grant that governs a thing is the grant for that thing, wherever the button happens to live.

What happens to a message when it arrives

Worth knowing, because it explains what you will see in the inbox.

- 1 **The number is normalised** to +91.... A number stored one way and a customer stored another would be two records for one buyer.
- 2 **The provider's message id is checked.** Twilio redelivers when a reply is slow or a deploy drops a request. A redelivery is discarded, not recorded twice.
- 3 **The customer book is searched** — the customer's own mobile and WhatsApp numbers, *and* every named contact's. The merchandiser who actually messages is usually not the person the account was opened under.
- 4 **Then open leads**, only if no customer matched. A customer always wins: a buyer is routinely both an account and a stale lead from before they were one.
- 5 **It is assigned.** A known customer goes to the account owner. An open lead goes to whoever is working it. An unknown number goes round-robin across marketing [§41.3].
- 6 **A second message from the same number joins the same conversation.** Never a new row.

Nothing creates a lead or an enquiry by itself

A message is not a qualified requirement — somebody says “hi”, somebody asks whether the plant is open on Saturday. Converting is an action a person takes once the conversation has turned into a real enquiry.

When it does not work

Check Twilio's own log first: **Monitor** → **Logs** → **Errors**, and **Messaging** → **Logs**. Twilio records every webhook attempt and the response it got.

What Twilio shows	What it means
11200 HTTP retrieval failure	Twilio could not reach the URL. Check HTTPS, the domain, and that the server is up.
401	Wrong or missing token. Compare the ?token= in the console against .env, exactly.
503	WHATSAPP_WEBHOOK_TOKEN is not set on the server, or the service was not restarted after setting it.
200, but no conversation	The message was accepted and refused — the response body says why.

A 200 with nothing in the inbox

The response body says why. The commonest is {"outcome": "rejected", "why": "no usable sender number"}, which means From did not arrive — almost always a body-parsing problem behind a proxy that rewrites the content type. {"outcome": "duplicate"} means the message id was already recorded, which is correct behaviour for a retry.

The conversation is there but assigned to nobody

Nobody is in the rotation. Check that at least one active user has department: marketing. The conversation is safe — it sits in the **Unassigned** queue rather than being dropped — but nobody is being told about it.

A known customer came in as unknown

Their number is not on the customer record in a form that matches. Open the conversation, link the customer, and the number is written onto that customer as you do it — so the next message from them matches by itself.

Security notes

The webhook URL is a credential. It carries the token, and anybody who has it can post messages into the inbox. Rotate it by changing WHATSAPP_WEBHOOK_TOKEN, restarting, then updating the Twilio console — in that order, accepting a minute of rejected messages in between.

Known limits, worth deciding about before go-live

Twilio's own signature is not yet checked. Twilio signs every webhook with an X-Twilio-Signature header computed from the URL, the body and your auth token. Validating it is strictly stronger than a shared secret — it cannot be replayed against a different payload and does not sit in a URL. The server does not do this today; it is a worthwhile follow-up before the volume gets interesting.

Media URLs are stored, not the media. A photo a customer sends is kept as a Twilio URL; those need the account credentials to fetch and do not live forever. If artwork received over WhatsApp must be kept, it has to be downloaded into the attachment store — that is not built yet.

The outbound side (§42)

Already built, and it shares this account. Two things differ:

- **Templates.** Outside a 24-hour window from the customer's last message, Meta only allows pre-approved template messages — free text is refused with error 63016. Approve templates in the Twilio console and put their SIDs in `WHATSAPP_TEMPLATE_*` in `.env`.
- **A person sends every message.** An internal status change never reaches a customer by itself [§42]. Somebody picks the update, previews the draft, edits it, and confirms.

Quick reference

```
# .env
TWILIO_ACCOUNT_SID=
TWILIO_AUTH_TOKEN=
TWILIO_WHATSAPP_FROM=whatsapp:+...
WHATSAPP_WEBHOOK_TOKEN=

# Twilio webhook ("When a message comes in", HTTP POST)
https://your-domain.com/api/whatsapp/inbound?token=...

# The module grant marketing needs
whatsapp          # read to see the inbox, write to act on it
enquiries: write   # additionally, to convert a conversation
```

Endpoints

Endpoint	What it does
POST /api/whatsapp/inbound	The webhook. Token-guarded, no login.
GET /api/whatsapp/threads	The inbox. ?open=true, ?status=, ?unassigned=true.
GET /api/whatsapp/threads/:id	One conversation, with its messages.
POST /api/whatsapp/threads/:id/read	Marks it read.
PATCH /api/whatsapp/threads/:id	Assign, set the queue, link a customer or lead, add a note.
POST /api/whatsapp/threads/:id/enquiry	Converts it. Supply the requirement; everything else comes off the thread.

The queues [§41.5]

Queue	Meaning
new	Arrived, nobody has worked it yet.
waiting_for_customer	We have replied; a new message reopens it as New.
sample_requested	A sample is with the buyer.
pricing_required	Waiting on a costing.
converted	An enquiry has been raised. Set by converting, never typed.
closed	Finished with. Drops out of the working inbox.

This guide is maintained in the server repository at docs/WHATSAPP-SETUP.md. If the setup changes, change it there — the PDF is a render of that file.